

Name: Voot Tangkaratt
Email: voot.tangkaratt@gmail.com
Github page: <https://voot-t.github.io>
Location: Tokyo, Japan

I am a machine learning researcher focusing on **robust imitation and reinforcement learning for robotics**. My research goal is to make smart robots that are useful to everybody. I am originally from Thailand and have been living in Japan since 2012.

Technical interests:

- Robust imitation and reinforcement learning for robot manipulators and humanoids.
- Learning algorithms for multi-tasks and quick adaptation to unseen tasks (lifelong and few-shot learning).

Work history:

- **Senior Research Scientist** at Sony AI, January 2022 - present
 - **Responsibility:** Conduct research on **vision-and-tactile-based robot learning algorithms for fine manipulation in gastronomy domains**.
- **Postdoc Researcher** at RIKEN AIP, April 2017 – December 2021
 - **Responsibility:** Conduct research on **robust machine learning algorithms under real-world data constraints**.

Education:

- **PhD in Computer Science**, Graduate School of Information Science and Technology, University of Tokyo, Japan, 2017
 - **Thesis:** Single-step Dimension Reduction for High-dimensional Data Analysis with Application in Reinforcement Learning
- **Masters in Computer Science**, Tokyo Institute of Technology, Japan, 2014
 - **Thesis:** Least-Squares Conditional Density Estimation with Dimensionality Reduction
- **Bachelors in Engineering**, Chulalongkorn University, Thailand, 2011

Technical skills:

- **Machine learning (ML) skills:**

- Expertised in **vision-based robust imitation and reinforcement learning**.
- Experience with general **robust ML**, such as robust objective and optimization, data augmentation, domain randomization, and uncertainty estimation.
- Experience with **reward design and learning from human experts**, including industry experts who are not familiar with AI/ML or robotics.
- Experience in **deep learning architectures** (e.g., MLP, RNN, Convolutions, Transformers) and shallow architectures (e.g., non-parametric, kernels).

- **Robot learning skills:**

- Experience with using **real-world multi-modal data from proprioceptor, vision, and tactile sensors** collected from human demonstrators.
- Experience with using robust ML to **address real-world sensor noises and sim-to-real gaps** in robot learning.
- Experience with **prototyping a robot learning pipeline on real robots**:
 - The pipeline includes collecting raw data, processing data, training models on local and GPU clusters, and deploying models on real robots.
- Familiarity with using **position and impedance controllers** in real robots.

- **Programming skills:**

- Experience in **Python** programming.
- Experience in **deep learning libraries** (Pytorch, Jax, Keras, and Tensorflow).
- Experience in using open-source **vision and vision-language-action** models for prototyping and evaluation.
- Experience with **Robot Operating System (ROS)** 1 and 2.
- Experience with **robot simulation libraries** (Gymnasium, Mujoco, and Pybullet).
- Familiarity with software development tools such as Git and Docker containers.

- **Language skills:**

- Academic-level English, Intermediate-level Japanese, and Native Thai.

Non-technical skills:

- Experience with managing a small group of members as a **sub-team leader**.
 - Responsibility includes managing roadmaps, milestones, meetings, software and hardware resources, multi-modal datasets, and large-scale GPU clusters of the subteam.
- Experience with **collaborating with external organizations** for research, development, publications, and patents.

Awards:

- RIKEN 2019 Ohbu Award (Research & Industry Partnership Incentive Award).
- Best student presentation award at Information-Based Induction Sciences and Machine Learning (IBIS) 2016.

Selected Publications:

- Robust Imitation Learning from Noisy Demonstrations
 - **Voot Tangkaratt**, Nontawat Charoenphakdee, Masashi Sugiyama
 - In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021
- Variational Imitation Learning with Diverse-quality Demonstrations
 - **Voot Tangkaratt**, Bo Han, Mohammad Emtiyaz Khan, Masashi Sugiyama
 - In *International Conference on Machine Learning (ICML)*, 2020
- Simultaneous Planning for Item Picking and Placing by Deep Reinforcement Learning
 - Tatsuya Tanaka, Toshimitsu Kaneko, Masahiro Sekine, **Voot Tangkaratt**, Masashi Sugiyama
 - In *International Conference on Intelligent Robots and Systems (IROS)*, 2020
- Imitation Learning from Imperfect Demonstration
 - Yueh-Hua Wu, Nontawat Charoenphakdee, Han Bao, **Voot Tangkaratt**, Masashi Sugiyama
 - In *International Conference on Machine Learning (ICML)*, 2019

- TD-regularized actor-critic methods
 - Simone Parisi, **Voot Tangkaratt**, Jan Peters, Mohammad Emtiyaz Khan
 - In *Machine Learning journal*, 2019
- Fast and Scalable Bayesian Deep Learning by Weight-perturbation in Adam
 - Mohammad Khan, Didrik Nielsen, **Voot Tangkaratt**, Wu Lin, Yarin Gal, Akash Srivastava
 - In *International Conference on Machine Learning (ICML)*, 2018
- Guide actor-critic for continuous control
 - **Voot Tangkaratt**, Abbas Abdolmaleki, Masashi Sugiyama
 - In *International Conference on Learning Representations (ICLR)*, 2018

For a full publication list, please visit my [github page](#) or [google scholar](#).